Prathamesh Gite | Python Developer – Data Scientist & Full Stack

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Profile Summary

Python Developer with expertise in **QGIS and geospatial automation, specializing in high-performance plugins and custom GIS tools**. Experienced in **computer vision, machine learning, and deep learning, including fine-tuning SAM, Grounding DINO, and YOLO for object detection**. Skilled in developing scalable data pipelines with Python**, FastAPI, AWS, Docker**, processing **geospatial, aerial, and thermographic data** for actionable insights. Proficient in data preprocessing, feature engineering, model evaluation, and **data orchestration using Prefect for automating end-to-end workflows**. Adept at delivering reliable GIS tools, automated reporting systems, and dashboards that support accurate, data-driven decision-making.

Experience - Datasee.AI (Dec 2023 – present)

- Developed and optimized a **QGIS Plugin** using **KD-Tree and R-Tree** spatial indexing for rapid nearest-point lookup and polygon snapping, enabling **5–10× faster** large-scale geospatial processing and reducing manual GIS workload.
- Designed **computer vision and machine learning pipelines** to detect and classify solar panel defects, improving detection accuracy and reducing inspection time by **40–60%**.
- Fine-tuned **SAM, Grounding DINO, YOLO**, and other deep learning models for prompt-based object detection and garbage identification across diverse datasets, achieving **high-precision, low-latency** inference.
- Executed full **data science workflows** including data preprocessing, feature engineering, model training, validation, and performance optimization for geospatial and aerial imagery datasets.
- Built and deployed a **FastAPI backend** for seamless API integration, ensuring efficient communication between frontend applications and **AWS Cloud** services.
- Utilized **SQL databases** for structured data management, improving query performance and scalability across analytics and production systems.
- Developed automated reporting systems using **Python, Jinja2, HTML, and CSS**, reducing manual reporting time by **80%** and ensuring consistent, high-quality outputs.
- Containerized services using **Docker**, streamlining deployment pipelines and ensuring environment consistency across development and production.
- Processed, analyzed, and visualized **geospatial, aerial, and thermographic data** to support data-driven insights and decision-making for solar infrastructure projects.
- **Implemented data orchestration using Prefect:** designed a workflow where the application monitors local directories for project input data downloaded from **AWS S3**, triggers Prefect flows upon data availability, and executes a sequence of processing scripts in the correct order—ensuring fully automated, fault-tolerant batch processing.

Technical Skills

- Programming / Engineering Languages:
  Python, SQL, HTML, CSS

- Frameworks & Libraries:
  FastAPI, PyTorch, TensorFlow, OpenCV, NumPy, Pandas, sklearn

- Tools & Platforms:
  Docker, AWS (S3, Lambda), QGIS, Git, Jinja, Git Hub

- Other Skills:
  Computer Vision, Geospatial Data Processing, API Development, Data Visualization, SAM, Grounding Dino

Education

Program	University	% / CGPA	Year of Completion
B.E (Mechanical Engineering)	Pune University	8.15	2023
Diploma (Mechanical Engineering)	Maharashtra State Board of Technical Education, Mumbai.	86.00 %	2020

Certifications

- Computer Vision and Deep Learning – Udemy
- Advanced Artificial Intelligence and Machine Learning – Intellipaat & IIT Roorkee (Ongoing)

Language Skills

- Native language:
  Marathi

- Other:
  English, Hindi